Introduction to Basic and Advanced Statistical Modelling

Introduction

Alexandre Courtiol

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About this workshop

Why this workshop?

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- Many local actors collect invaluable data on biodiversity in one of the most biodiversity-rich regions in the world
- Much of these data remain unexploited and yet they could benefit both local conservation and the global scientific community
- We believe one reason is that many local actors have limited quantitative skills
- This workshop aims at strengthening local capacity by training local actors

The workshop structure

Module A: background knowledge to prepare for Module B

- foundation of the R language
- introduction to reproducible reporting workflows
- review of core statistical concepts
- introduction to Linear Models (LMs, GLMs)

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Module B: applied methods

- single-species occupancy models
- community-level frameworks
- predicting and mapping occurrence and richness across the study area

About me, you & us

The lecturers

Module A: Alexandre Courtiol

Module B: Rahel Sollmann

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- lecturer in data science & statistics

Some examples of the research I contributed to

SCIENCE ADVANCES | RESEARCH ARTICLE

EVOLUTIONARY BIOLOGY

The legacy of privilege: Social inheritance reverses sex differences in reproductive inequality in the spotted hyena

LETTER OPEN ACCESS ECOLOGY LETTERS

Shorter and Warmer Winters Expand the Hibernation Area of Bats in Europe

nature sustainability

Analysis https://doi.org/10.1038/s41893-022-00970-0

Protected area personnel and ranger numbers are insufficient to deliver global expectations

Science

REPORT | ADAPTATION

Genetic variance in fitness indicates rapid contemporary adaptive evolution in wild animals

nature COMMUNICATIONS

https://doi.org/10.1038/s41467-022-30366-9 OPEN

Mothers with higher twinning propensity had lower fertility in pre-industrial Europe

ORIGINAL ARTICLE MOLECULAR ECOLOGY WILEY

Sex-specific spatial variation in fitness in the highly dimorphic *Leucadendron rubrum*

Remote Sensing in Ecology and Conservation Open Access ZSL LET'S WORK FOR WILDLIFE

ORIGINAL RESEARCH

Assessing analytical methods for detecting spatiotemporal interactions between species from camera trapping data

nature ecology & evolution ARTICLES https://doi.org/10.1038/s41559-018-0718-9

Social support drives female dominance in the spotted hyaena

nature reviews genetics

Review Article | Published: 09 May 2018

The transition to modernity and chronic disease: mismatch and natural selection

nature COMMUNICATIONS

DOI: 10.1038/s41467-018-05515-8 OPEN

Differences in age-specific mortality between wild-caught and captive-born Asian elephants

Methods in Ecology and Evolution BRITISH ECOLOGICAL SOCIETY

Methods in Ecology and Evolution 2016, 7, 1457–1462 doi: 10.1111/2041-210X.12600

APPLICATION

`camtrapR`: **an R package for efficient camera trap data management**

SCIENCE ADVANCES | RESEARCH ARTICLE

CONSERVATION ECOLOGY

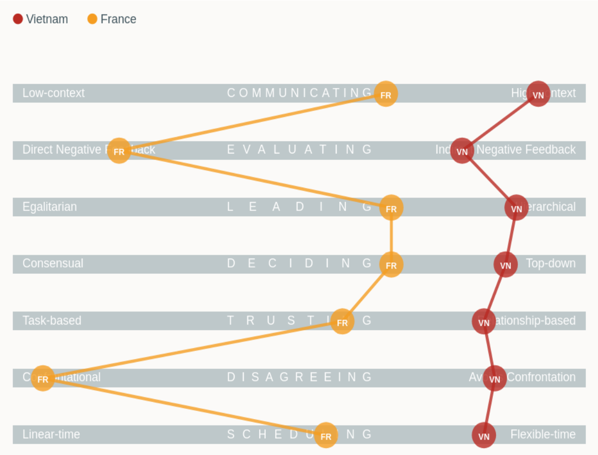
Planning tiger recovery: Understanding intraspecific variation for effective conservation

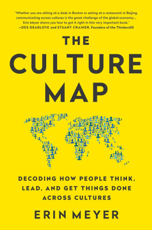
Who are you?

Workshop rules

- slides are provided, so invest your time in thinking (not just writing)
  $\rightarrow$ github.com/courtiol/intro_modelling_workshop
- try to learn actively (ask questions)
- treat each other as equal
- there is no dumb question
- avoid using LLMs unless instructed otherwise

Warning: beware of the culture gap





Vietnam vs France across 7 dimensions introduced in The Culture Map book
data source: https://erinmeyer.com/tools/culture-map-premium

What next?

Tentative schedule

Day	Date	Morning	Afternoon
Day 1	Aug. 24	Introduction + R language	R language
Day 2	Aug. 25	Reproducible reporting + Practice	Core concepts
Day 3	Aug. 26	Core concepts	Linear Models
Day 4	Aug. 27	Linear Models	Linear Models

Questions?